

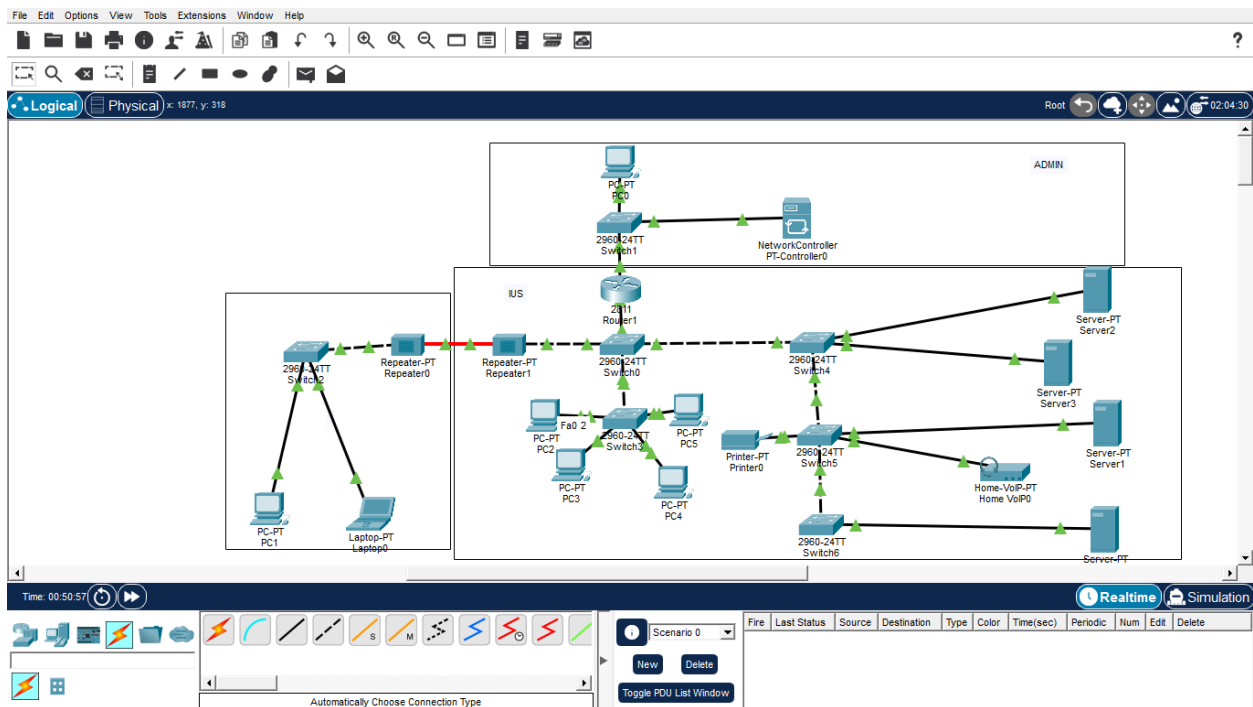
# INSTITUT UNIVERSITAIRE DES SCIENCES

TD7 ResauI

INNOCENT COFFY MANAHENN

L3

FST



Router1

Physical Config CLI Attributes

IOS Command Line Interface

```

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface f0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#ip dhcp pool ADMIN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool IUS
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.2.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.2
Router(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.2
Router(config)#
  
```

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Automatically Choose Connection Type

Toggle PDU List Window

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 8.8.8.8

IPv6 Configuration

☒ Automatic ☐ Static IPv6 request successful.

IPv6 Address 2001:DB8:2:0:BAED:D7BF:2E38:AF87 / 64

Link Local Address FE80::230:F2FF:FE99:9E0B

Default Gateway FE80::202:16FF:FEC9:4D02

DNS Server 2001:4860:4860:8888

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

Automatically Choose Connection Type

Toggle PDU List Window

PC5

Physical Config Desktop Programming Attributes

Command Prompt

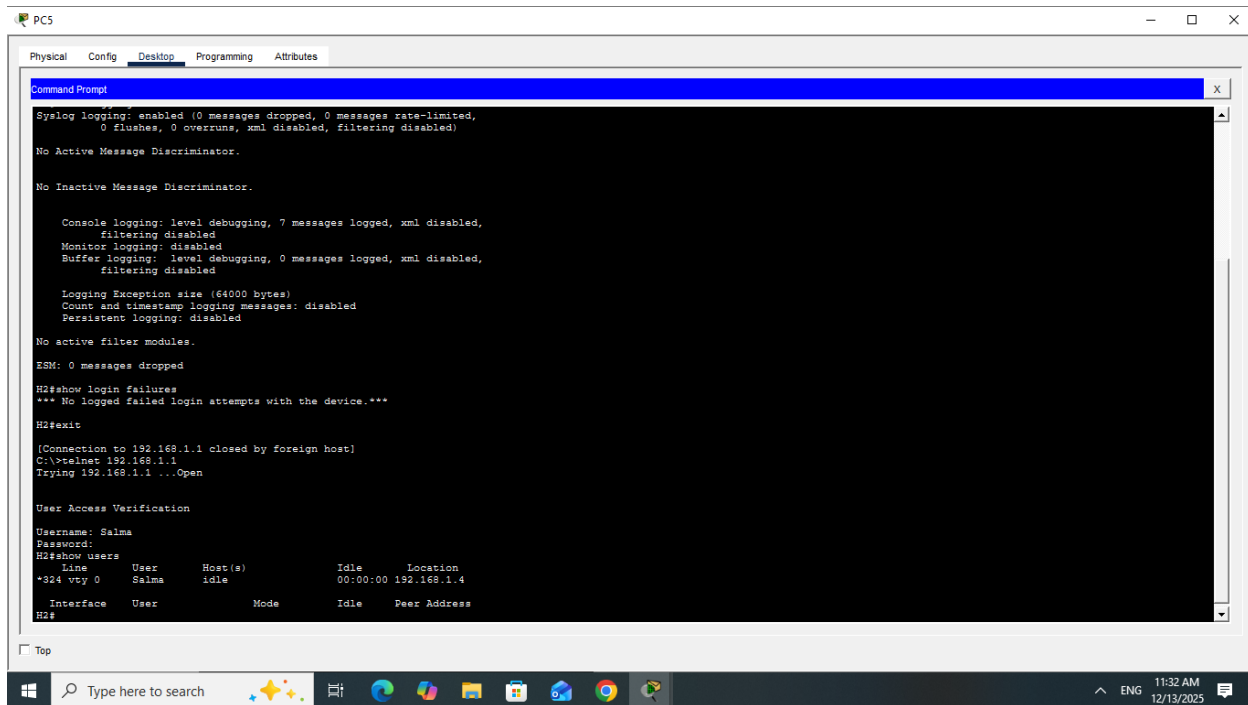
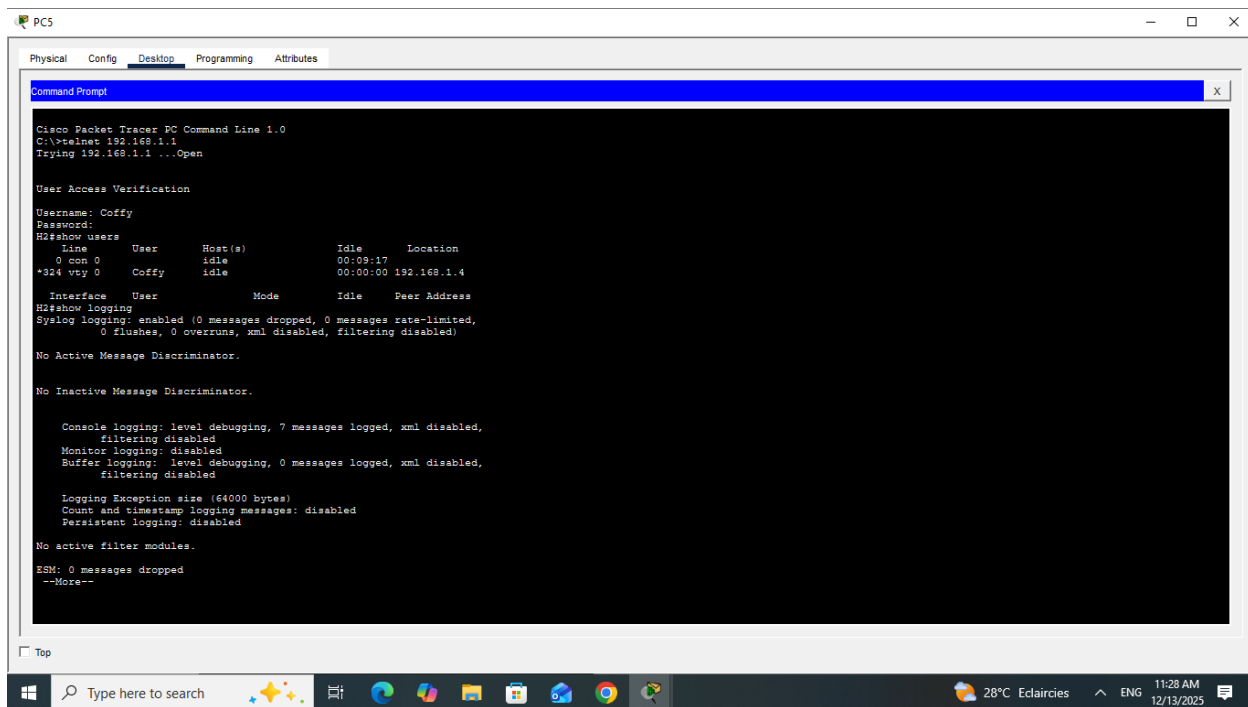
```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

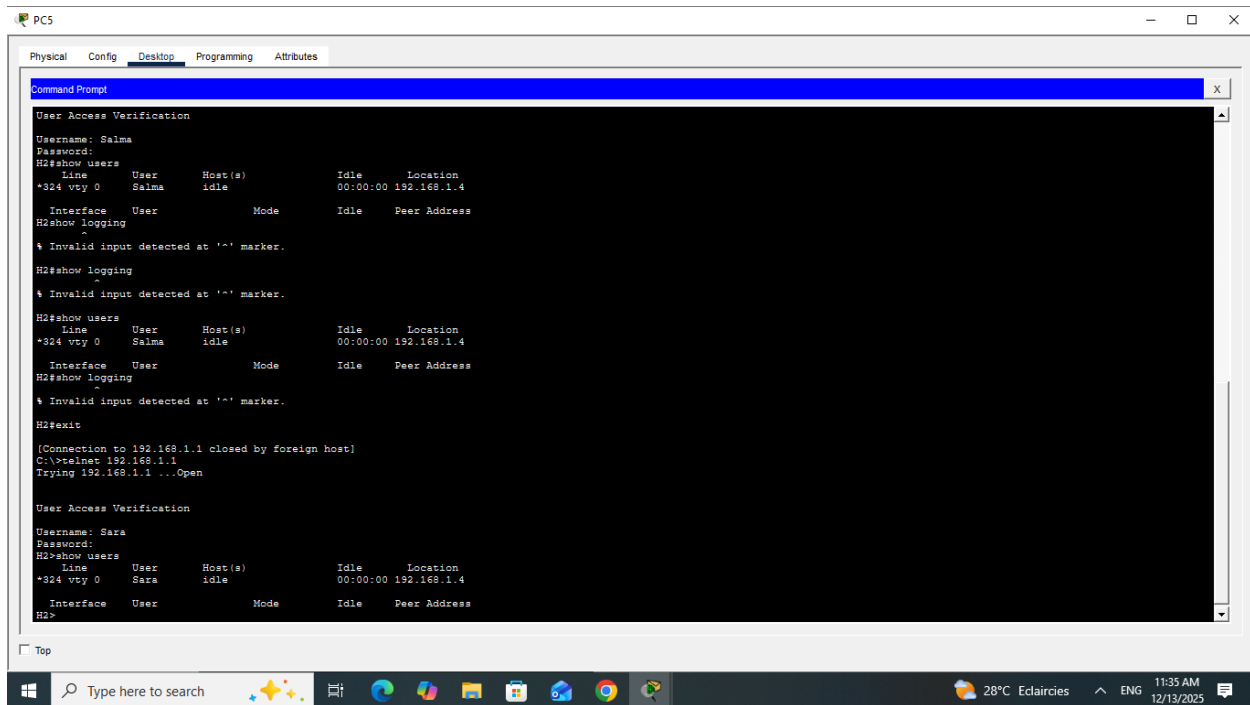
User Access Verification
Username: Coffy
Password:
H2#show users
Line          User             Host(s)          Idle           Location
* 0 con 0
+* 324 vty 0    Coffy            idle            00:09:17      192.168.1.4
Interface     User             Mode             Idle           Peer Address
H2#
```

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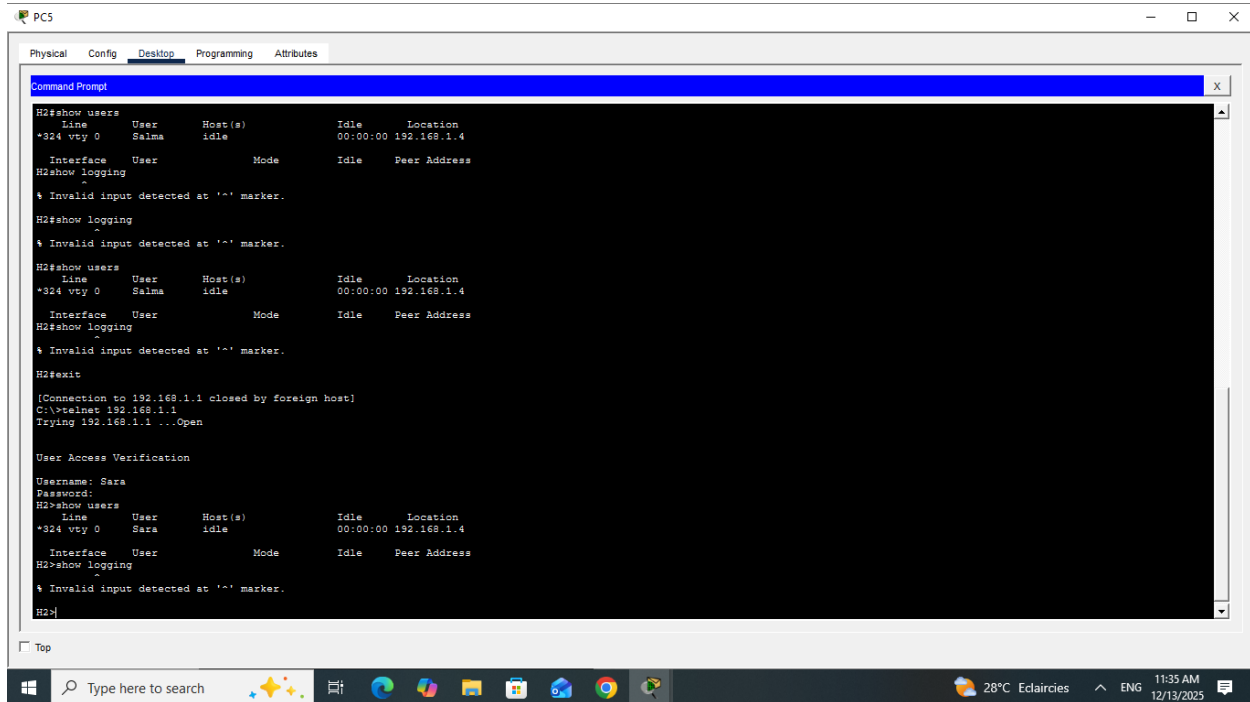




The screenshot shows a Windows desktop environment with a window titled 'PCS'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the following sequence of commands and outputs:

```
User Access Verification
Username: Salma
Password:
H2#show users
  Line    User    Host(s)    Idle    Location
*324 vty 0    Salma      idle     00:00:00 192.168.1.4
  Interface  User      Mode      Idle    Peer Address
H2#show logging
^
* Invalid input detected at '^' marker.
H2#show logging
^
* Invalid input detected at '^' marker.
H2#show users
  Line    User    Host(s)    Idle    Location
*324 vty 0    Salma      idle     00:00:00 192.168.1.4
  Interface  User      Mode      Idle    Peer Address
H2#show logging
^
* Invalid input detected at '^' marker.
H2#exit
[Connection to 192.168.1.1 closed by foreign host]
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification
Username: Sara
Password:
H2#show users
  Line    User    Host(s)    Idle    Location
*324 vty 0    Sara      idle     00:00:00 192.168.1.4
  Interface  User      Mode      Idle    Peer Address
H2#
```



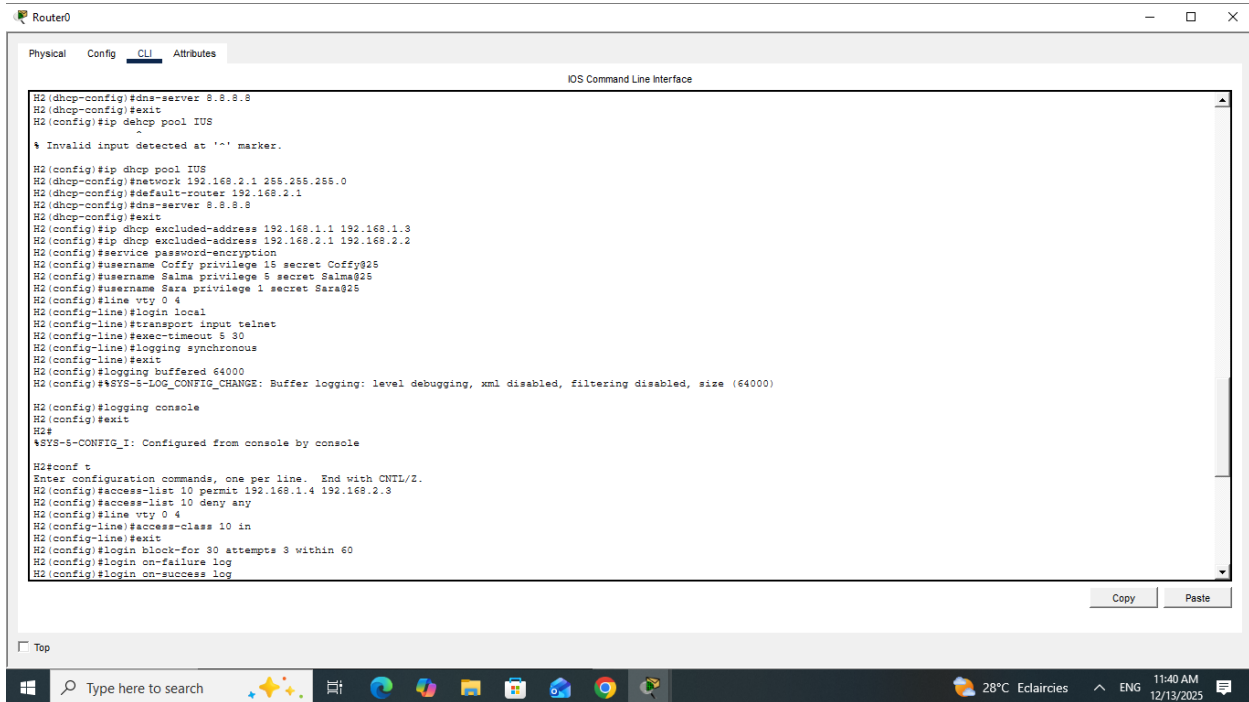
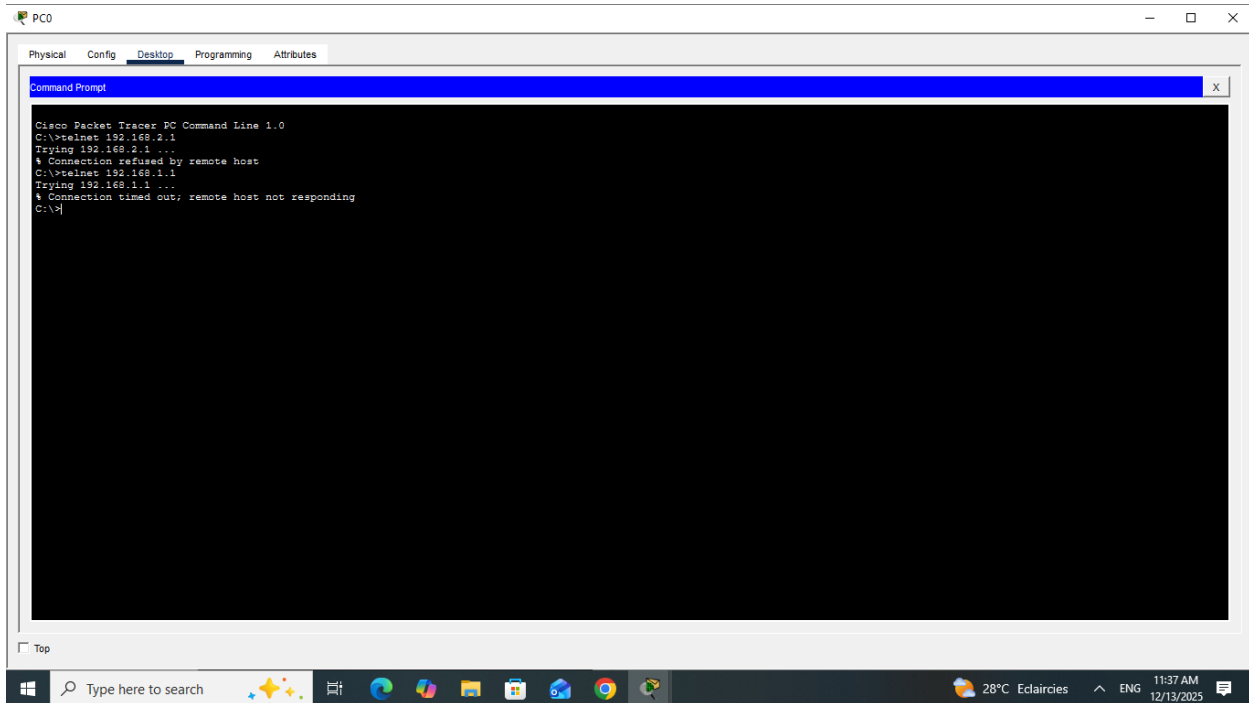
This screenshot is identical to the one above, showing the same 'PCS' window and 'Command Prompt' output. The sequence of commands and outputs is the same, ending with the prompt 'H2#'.

## Reponses

1)telnet est considere comme non securise parce qu'il ne chiffre aucune donnee echange entre le client et le serveur

2)avec telnet toutes ces donnees circulent sans chiffrement :nom d'utilisateur,mots de passe,lescommandes qu'a utilise l'utilisateur,et reponses et resultats sur le serveur, aussi les configuratio du système

3)telnet est deconseille car il y a une absence totale de chiffrement, intercepter des donnees est tres faciles. Il n'y a aucune garantie d'authenticite du serveur et aussi car telnet n'est pas conforme aux normes de securites (iso RGPD bonnes cybersecurite)



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
H2(config-if)#exit
H2(config)#int f0/1
H2(config-if)#ip address 192.168.2.1 255.255.255.0
H2(config-if)#no shutdown

H2(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

H2(config-if)#exit
H2(config)#ip dhcp pool ADMIN
H2(dhcp-config)#network 192.168.1.1 255.255.255.0
H2(dhcp-config)#default-router 192.168.1.3
H2(dhcp-config)#dns-server 8.8.8.8
H2(dhcp-config)#exit
H2(config)#ip dhcp pool IUS
H2(dhcp-config)#network 192.168.2.1 255.255.255.0
H2(dhcp-config)#default-router 192.168.2.1
H2(dhcp-config)#dns-server 8.8.8.8
H2(dhcp-config)#exit
H2(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.3
H2(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.2
H2(config)#service password-encryption
H2(config)#username Coffy privilege 15 secret Coffy@25
H2(config)#username Salma privilege 5 secret Salma@25
H2(config)#username Sara privilege 1 secret Sara@25
H2(config)#line vty 0 4
H2(config-line)#login local
H2(config-line)#transport input telnet
H2(config-line)#exec-timeout 5 30
H2(config-line)#logging synchronous
H2(config-line)#exit
H2(config)#logging buffered 64000
H2(config)#SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, filtering disabled, size (64000)
H2(config)#logging console
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname H2
H2(config)#int f0/0
H2(config-if)#ip address 192.168.1.1 255.255.255.0
H2(config-if)#no shutdown

H2(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

H2(config-if)#exit
H2(config)#int f0/1
H2(config-if)#ip address 192.168.2.1 255.255.255.0
H2(config-if)#no shutdown

H2(config-if)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

H2(config-if)#exit
H2(config)#ip dhcp pool ADMIN
H2(dhcp-config)#network 192.168.1.1 255.255.255.0
H2(dhcp-config)#default-router 192.168.1.3
H2(dhcp-config)#dns-server 8.8.8.8
H2(dhcp-config)#exit
H2(config)#ip dhcp pool IUS
H2(dhcp-config)#network 192.168.2.1 255.255.255.0
H2(dhcp-config)#default-router 192.168.2.1
H2(dhcp-config)#dns-server 8.8.8.8
H2(dhcp-config)#exit
```

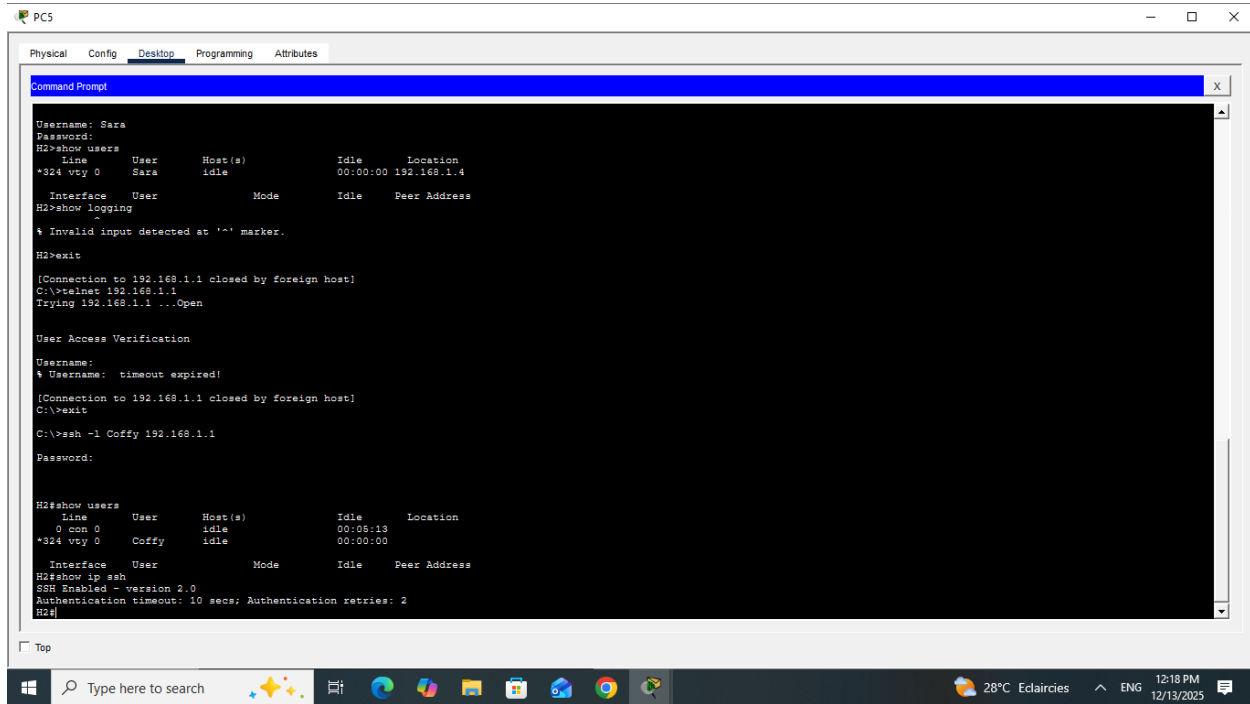
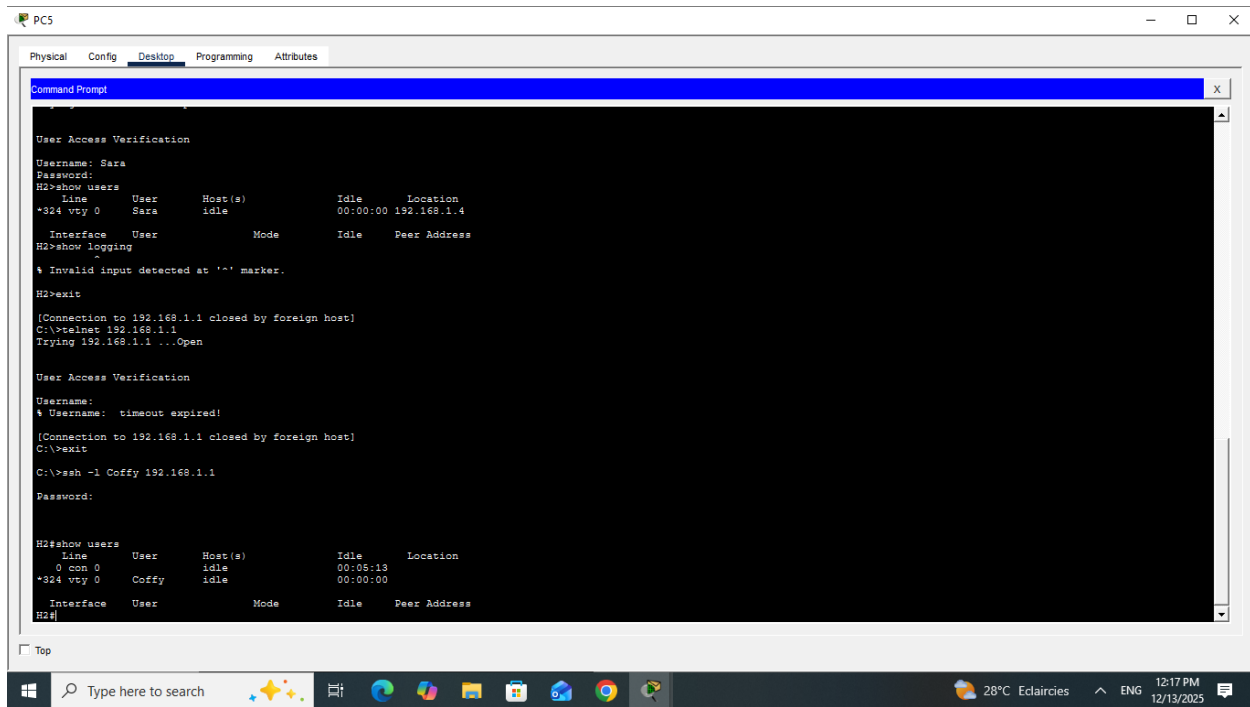
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PCS

Physical Config Desktop Programming Attributes

Command Prompt

User Access Verification  
Username:  
\* Username: timeout expired!  
[Connection to 192.168.1.1 closed by foreign host]  
C:\>exit  
C:\>ssh -l Coffy 192.168.1.1  
Password:  
  
H2#show users  
Line User Host(s) Idle Location  
0 con 0 idle 00:05:13  
\*324 vty 0 Coffy idle 00:00:00  
  
Interface User Mode Idle Peer Address  
H2#show ip ssh  
SSH Enabled - version 2.0  
Authentication timeout: 10 secs; Authentication retries: 2  
H2#show crypto key mypubkey rsa  
\* Key pair was generated at: 1:20:43 UTC March 1 1993  
Key name: H2.ius.edu  
Storage Device: not specified  
Usage: General Purpose Key  
Key is not exportable.  
Key Data:  
00001a8e 000041d0 00005957 0000029d 000021a7 000074a5 0000094a 000008ad  
00001002 00007138 000010e4 00002d22 00005714 00004beb 00000952 00006189  
00003b0a 00000dc8 000026c1 00005b6b 0000651a 00001db9 00004cc4 671e  
\* Key pair was generated at: 1:20:43 UTC March 1 1993  
Key name: H2.ius.edu.server  
Temporary key  
Usage: Encryption Key  
Key is not exportable.  
Key Data:  
00004b0c 000049fb 000041f6 00005ce1 00001ffa 000005d0 000046e0 00006c29  
00004254 00002f74 0000141d 00003135 00004b26 00007253 000003bb 000048e8  
0000229e 00001979 000069e5 000056d8 00003dc9 0000664a 000060bf 31a3  
H2#

Top

Windows Taskbar

PCS

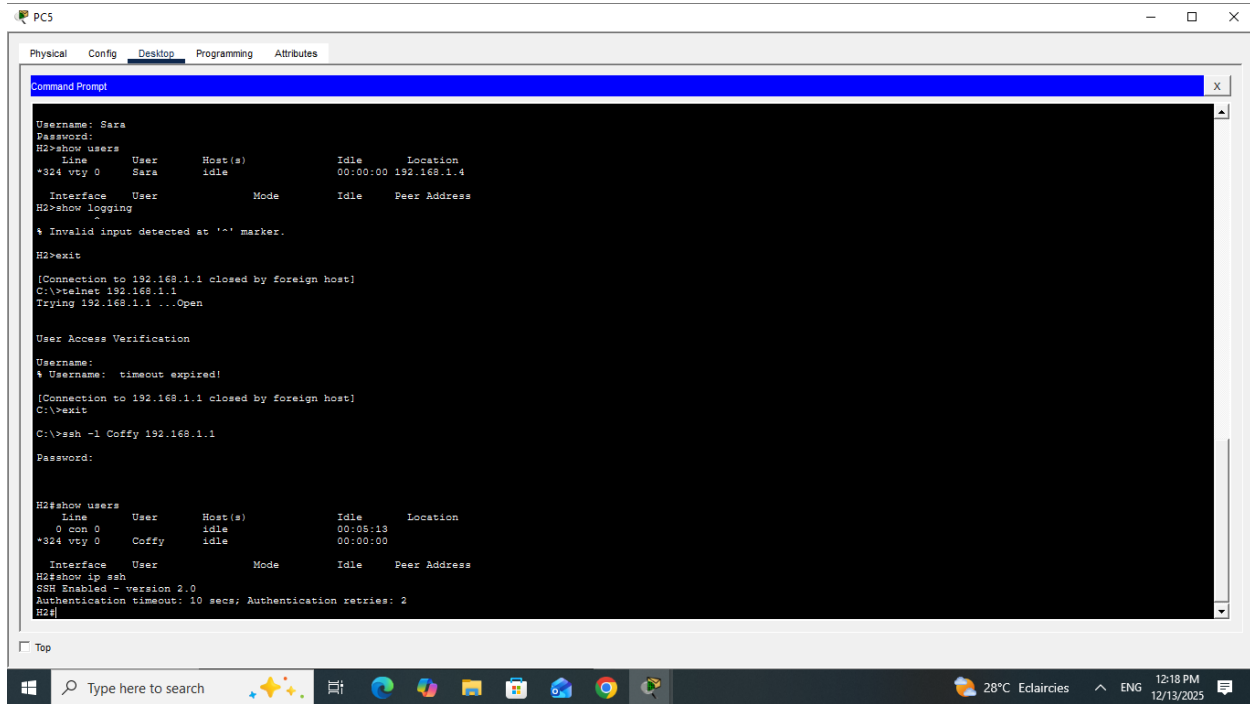
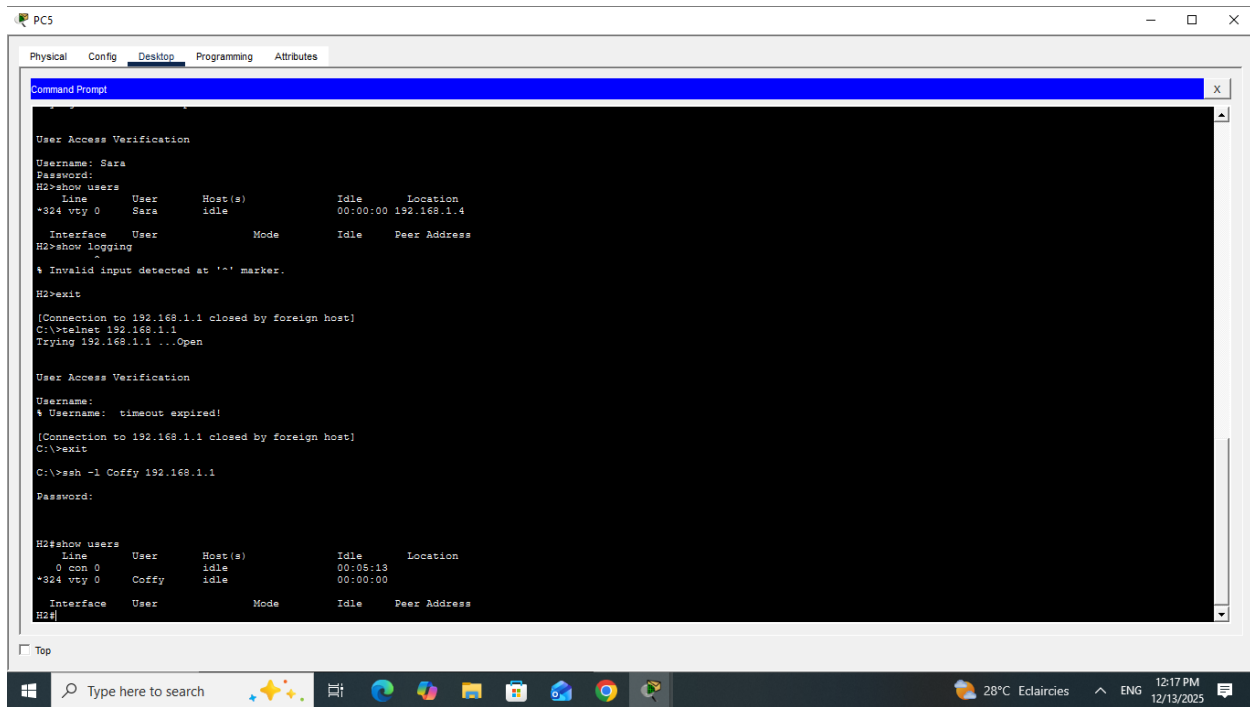
Physical Config Desktop Programming Attributes

Command Prompt

\* Username: timeout expired!  
[Connection to 192.168.1.1 closed by foreign host]  
C:\>exit  
C:\>ssh -l Coffy 192.168.1.1  
Password:  
  
H2#show users  
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00004254 00002f74 0000141d 00003135 00004b26 00007253 000003bb 000048e8  
0000229e 00001979 000069e5 000056d8 00003dc9 0000664a 000060bf 31a3  
H2#  
[Connection to 192.168.1.1 closed by foreign host]  
C:\>

Top

Windows Taskbar



PCS

Physical Config Desktop Programming Attributes

Command Prompt

User Access Verification

Username:

\* Username: timeout expired!

[Connection to 192.168.1.1 closed by foreign host]

C:\>exit

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Password:

H2#show users

Line User Host(s) Idle Location

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\*324 vty 0 Coffy idle 00:00:00

Interface User Mode Idle Peer Address

H2#show ip ssh

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Authentication timeout: 10 secs; Authentication retries: 2

H2#show crypto key mypubkey rsa

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Key name: H2.ius.edu

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Key name: H2.ius.edu.server

Temporary key

Usage: Encryption Key

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Key Data:

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0000229e 00001979 000069e5 000056d8 00003dc9 0000664a 000060bf 31a3

H2#

Top

Type here to search

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ENG

12:19 PM

12/13/2025

PCS

Physical Config Desktop Programming Attributes

Command Prompt

\* Username: timeout expired!

[Connection to 192.168.1.1 closed by foreign host]

C:\>exit

C:\>ssh -l Coffy 192.168.1.1

Password:

H2#show users

Line User Host(s) Idle Location

0 con 0 idle 00:05:13

\*324 vty 0 Coffy idle 00:00:00

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00004254 00002f74 0000141d 00003135 00004b26 00007253 000003bb 000048e8

0000229e 00001979 000069e5 000056d8 00003dc9 0000664a 000060bf 31a3

H2#

[Connection to 192.168.1.1 closed by foreign host]

C:\>

Top

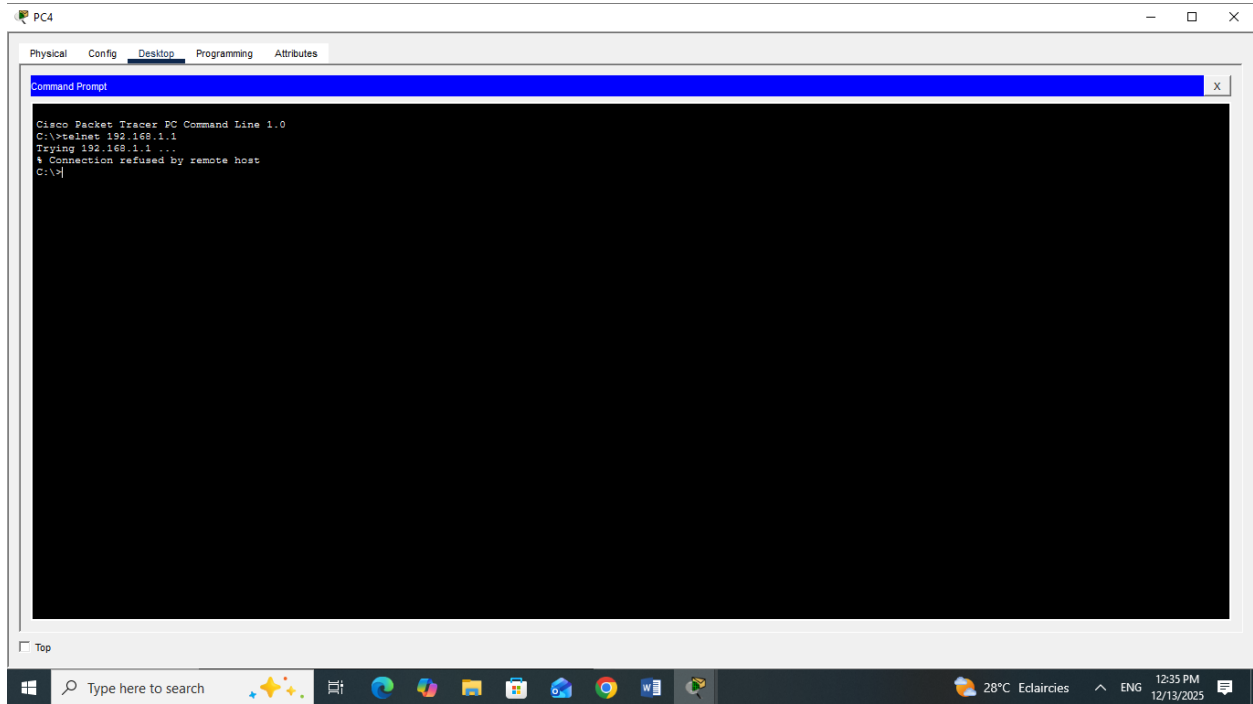
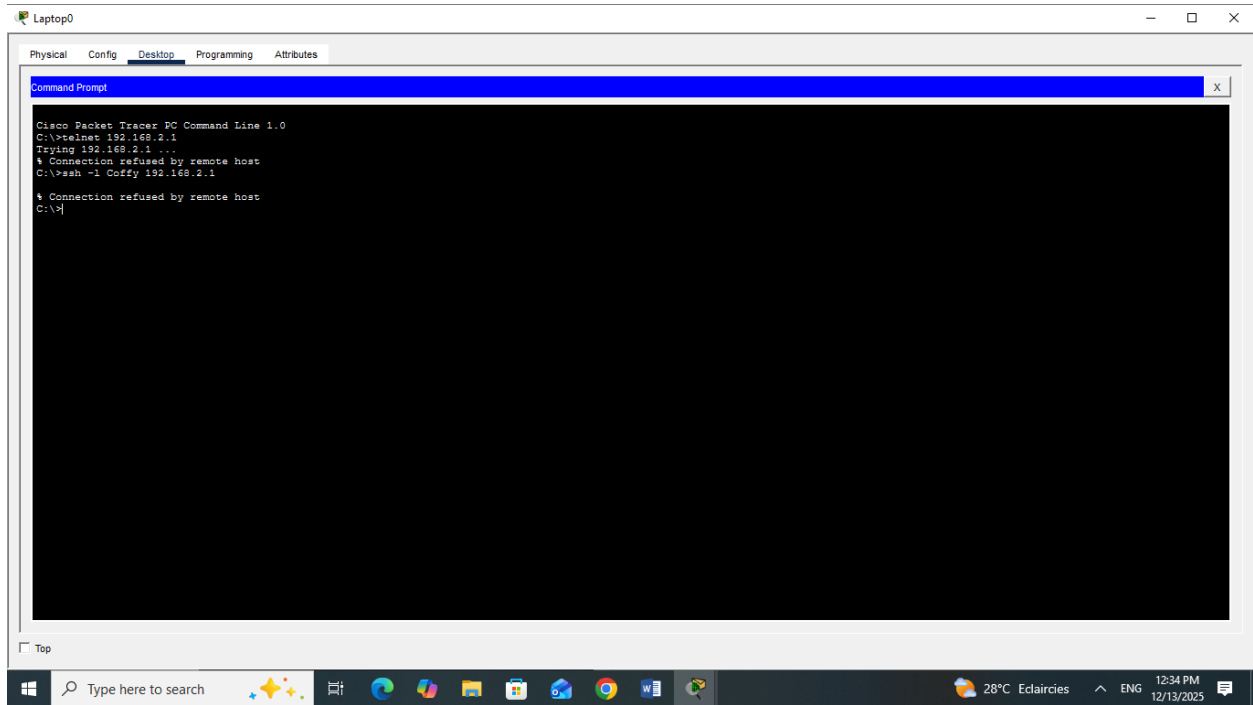
Type here to search

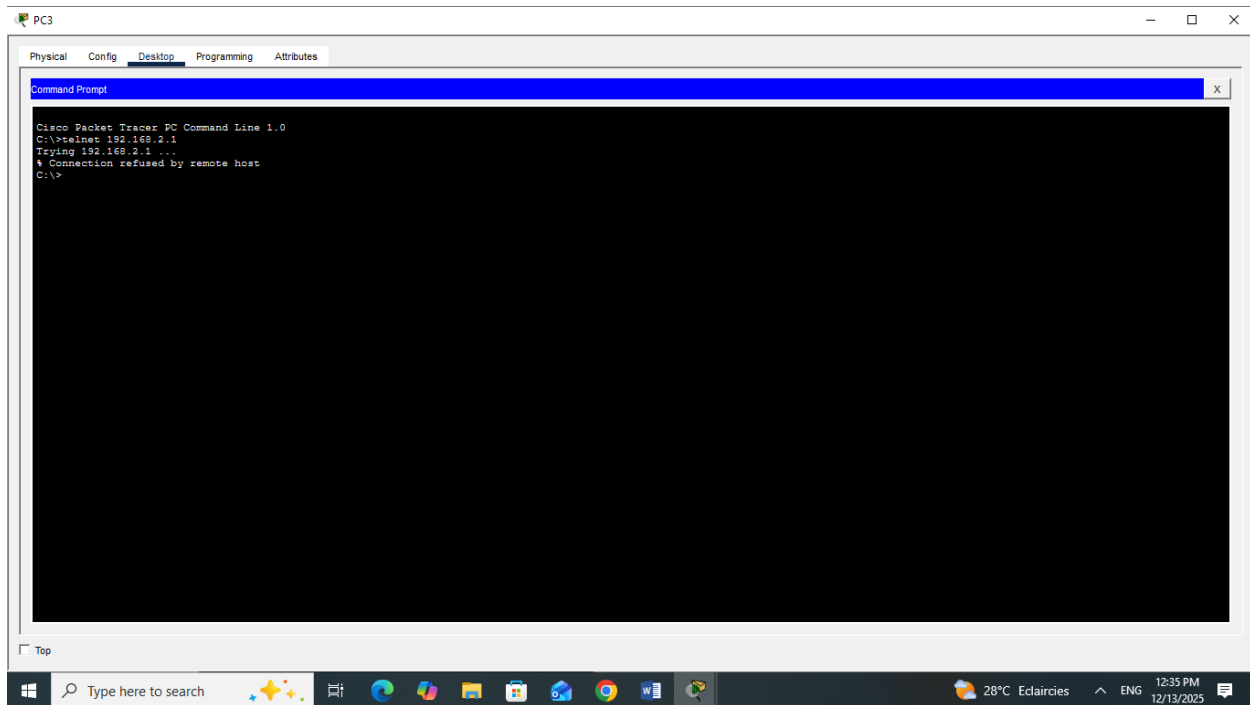
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ENG

12:24 PM

12/13/2025





## Réponses

- 1) SSH1 est une ancienne version du protocole tandis que SSHv2 est plus moderne et sécurisée, la ou SSH1 est vulnérable SSHv2 corrige les failles. Gestion faible de l'intégrité des données pour une forte vérification de l'intégrité du SSHv2 et aussi que SSHv2 peut supporter de nombreux algorithmes modernes.
- 2) RSA 1024 bits n'est plus recommandé car la clé est devenue trop courte face à la puissance des calculs actuels.
- 3) si on désactive le domaine local
  - a) les comptes du domaine ne pourront pas être authentifiés
  - b) les services dépendant du domaine cesseront de fonctionner
  - c) les stratégies de groupes (GPO) ne s'appliquent plus
  - d) pertes d'accès aux ressources partagées

en conclusion en production ssh est plus conforme que telnet pour la protection des donnees j'ai compris que telnet c'est du passe de l'administration alors que ssh represente le present et l'avenir et que dans tout environnement professionnel et critique telnet doit être évité et ssh privilégié